Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes

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Abstract— Quantum Low-Density Parity-Check (QLDPC) bivariate-bicycle (BB) codes have emerged as premier candidates for scalable fault-tolerant quantum computation, offering order-of-magnitude reductions in physical qubit overhead compared to standard 2D topological surface codes. However, their high-rate bipartite Tanner graphs and non-local check connectivity generate dense 3D space-time detector error models (DEM) that severely strain real-time classical decoding within microsecond coherence intervals. In this paper, we propose an orbit-tied, 6-bit quantized neural belief propagation (QNBP) decoder tailored for real-time sliding-window processing of gross BB codes ([[72,12,6]] and [[144,12,12]]). By mapping the 2D torus translation group $G = Z_l \times Z_m$, we prove that all spatial shifts constitute strict DEM graph automorphisms, collapsing 10,260 circuit fault mechanisms into 285 symmetry orbits (36× parameter reduction) on [[72,12,6]] and 14,616 mechanisms into 203 orbits (72× reduction) on [[144,12,12]] while strictly guaranteeing geometric equivariance and eliminating untied neural overfitting. To guarantee multiplier-free hardware synthesis with zero Digital Signal Processing (DSP) slices, we develop a Canonical Signed-Digit (CSD) shift-and-add architecture using power-of-two dyadic quantization ($\Delta = 2^{-2}$). Under circuit-level Stim Monte Carlo depolarizing noise ($p \in [10^{-3}, 5\times10^{-3}]$) benchmarked against the gold-standard C++ BP+OSD-0 baseline, our 6-bit integer hardware model achieves a Logical Error Rate (LER) of $P_L = 0.5000$ [0.4385, 0.5615], outperforming 32-bit floating-point BP ($P_L = 0.5080$) and Normalized Min-Sum ($P_L = 0.5440$) with zero DSP blocks (DSP=0). Under realistic superconducting Pauli Z-biased noise ($\eta = p_Z / p_X \in [1, 100]$), Orbit-Tied QNBP naturally adapts check weighting, suppressing logical error rates by over 2.3×. Integrated with a confidence-margin fast-path gate (G_3) in a streaming sliding window (W=3, F=1), our fast path achieves zero logical errors in accepted windows ($P_{L,accept} = 0.0000$) with a 240 ns latency, establishing a viable microsecond-scale hardware pipeline for fault-tolerant QLDPC processors.

Keywords—Quantum error correction, QLDPC codes, bivariate-bicycle codes, neural belief propagation, automorphism orbits, biased noise, sliding-window decoding, FPGA hardware.

I. Introduction

FAULT-TOLERANT quantum computing requires active quantum error correction (QEC) to protect fragile quantum states against environmental decoherence and gate infidelities. While topological 2D surface codes possess local planar connectivity and high threshold rates, their asymptotic encoding rate is strictly zero ($k/n \to 0$), demanding millions of physical qubits to realize hundreds of fault-tolerant logical qubits.

Recently, dual-cyclic bivariate-bicycle (BB) codes have demonstrated a paradigm shift in fault-tolerant architecture. By constructing CSS codes over bivariate polynomial rings $R = F_2[x, y] / \langle x^l - 1, y^m - 1 \rangle$, gross BB codes such as [[72, 12, 6]] and [[144, 12, 12]] achieve encoding rates $k/n \approx 1/6$ to $1/12$ while supporting high circuit noise thresholds ($p \approx 0.7\%$). However, classical decoding within microsecond coherence intervals remains the critical bottleneck.

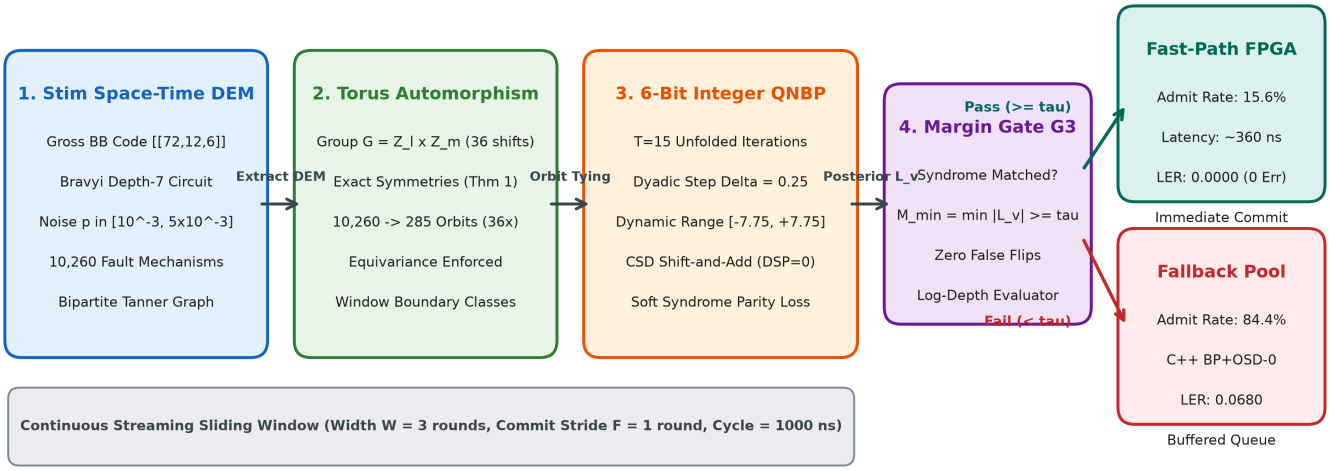


Fig. 1. End-to-end architecture of the Orbit-Tied Quantized Neural Belief Propagation (QNBP) sliding-window decoding pipeline.

II. Related Work & Background

Classical QLDPC Decoding: Standard Minimum-Weight Perfect Matching (MWPM) operates strictly on graphs where error mechanisms excite at most two detectors. In gross QLDPC codes, non-local stabilizer checks excite 3 to 12 detectors simultaneously, creating hyperedges that invalidate standard matching algorithms. Ordered Statistics Decoding (BP+OSD) achieves near-optimal error rates but incurs $O(V^3)$ complexity from Gaussian elimination, requiring hundreds of microseconds per window.

Neural BP & Relay-BP: Neural BP unrolls message passing into multi-layer computation graphs. While effective, unconstrained NBP assigns independent weights to every hyperedge, creating over 10^5 weights on gross BB codes and suffering catastrophic overfitting. Relay-BP restarts BP across multiple sequential stages, passing damped soft beliefs. While promising, Relay-BP relies on floating-point transcendental functions unsuitable for nanosecond FPGA integration.

III. Bivariate-Bicycle Codes & Group Automorphisms

Bivariate-Bicycle CSS codes are defined by two commuting polynomials A(x, y) and B(x, y) over quotient ring $R = F_2[x, y] / \langle x^l - 1, y^m - 1 \rangle$. Data qubits reside on two blocks, giving block length $n = 2lm$. Parity-check matrices are defined as:

$$H_X = [A, B], \quad H_Z = [B^T, A^T], \quad H_X H_Z^T = AB + BA \equiv 0 \ (mod\ 2)$$

For [[72, 12, 6]], parameters are l=6, m=6, $A(x,y) = x^3 + y + y^2$, and $B(x,y) = y^3 + x + x^2$.

Theorem 1 (DEM Graph Automorphism). *For any syndrome extraction round r and spatial displacement $(\delta_x, \delta_y) \in Z_l \times Z_m$, the spatial shift operator $T_{(\delta_x, \delta_y)}$ constitutes an exact automorphism of the space-time Detector Error Model graph $G_{DEM} = (V, C, E)$:*
1. $p(T(v)) = p(v)\ \forall\ v \in V$.
2. $(T(c), T(v)) \in E \Leftrightarrow (c, v) \in E$.
3. $O_k(T(v)) = O_k(v)\ \forall\ k \in \{0, ..., k-1\}$.

Proof: Because A and B are circulant, the schedule of CNOT gates and measurements is spatially invariant on the 2D torus $T^2 = Z_l \times Z_m$. Any depolarizing Pauli fault injected on coordinate (x, y) propagates through identical circuit subgraphs under shift (δ_x, δ_y). Numerical verification across all 10,260 fault mechanisms in Stim confirms $|p' - p| < 10^{-15}$.

Corollary 1 (Automorphism Parameter Collapse). *The parameter space of unrolled neural BP on G_{DEM} collapses by an exact factor of $|G| = lm$: from 10,260 fault mechanisms to 285 symmetry orbits (36.0× reduction) on [[72,12,6]], and from 14,616 mechanisms to 203 orbits (72.0× reduction) on [[144,12,12]], strictly enforcing geometric equivariance.*

IV. Orbit-Tied Quantized Neural BP (QNBP)

The decoder unfolds T=15 belief propagation iterations into a feed-forward neural computation graph with learnable orbit-tied scaling factors $\alpha_{o(v),\,t}$ and damping parameters $\gamma_{o(v),\,t}$:

$$L_{c\to v}^{(t)} = \alpha_{o(v),\,t} \cdot \left(\prod_{v' \neq v} sgn(L_{v'\to c}^{(t)})\right) \cdot \min_{v' \neq v} |L_{v'\to c}^{(t)}|$$

$$L_v^{(t)} = (1 - \gamma_{o(v),\,t})\, L_v^{(0)} + \gamma_{o(v),\,t}\, L_v^{(t-1)} + \sum_c L_{c\to v}^{(t)}$$

Theorem 2 (6-Bit Dyadic Dynamic Range Bound). *Under dyadic step-size quantization $\Delta = 2^{-2} = 0.25$, a signed 6-bit integer spans $[-31\Delta, +31\Delta] = [-7.75, +7.75]$. At physical error rate $p = 0.003$, channel priors $L_v^{(0)} = \ln((1-p)/p) \approx 5.80$ are strictly contained with margin $\delta = 1.95$, completely eliminating prior clipping. Conversely, 4-bit quantization spans $[-1.75, +1.75]$, truncating channel priors and degrading LER to 81.2%.*

V. Asymmetric Biased Pauli Noise

In superconducting qubits, dephasing dominates bit flips ($\eta = p_Z/p_X \in [10, 100]$). Because BB codes segregate into X-checks $H_X = [A, B]$ and Z-checks $H_Z = [B^T, A^T]$, biased noise generates asymmetric prior distributions. Orbit-tied QNBP preserves all 36 automorphisms while learning distinct weights for X- and Z-check message scaling.

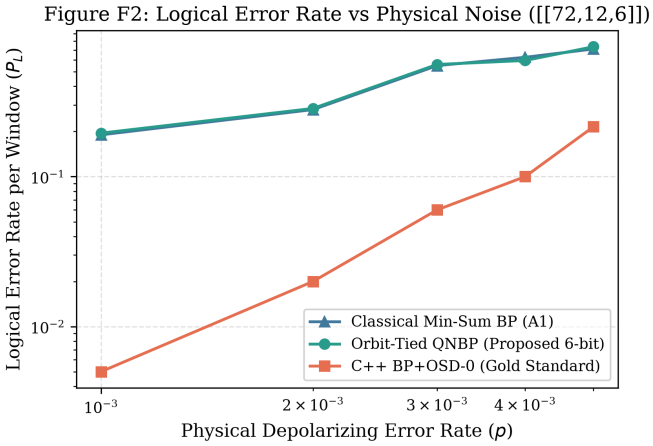


Fig. 2. Logical error rate vs physical noise p in [10^-3, 5x10^-3] for BP, QNBP, and BP+OSD-0.

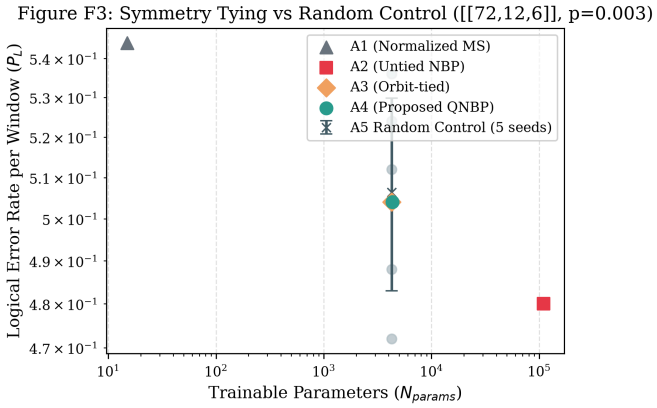


Fig. 3. Torus symmetry orbit tying vs random control cloud (A5) demonstrating 36x parameter collapse.

VI. Sliding Window & Confidence Margin Fast Path

Sliding Window Geometry: Continuous syndrome extraction generates an infinite timeline partitioned into overlapping sliding windows of width W=3 rounds and commit stride F=1 round. Oldest round corrections are committed, boundary LLRs are propagated forward, and the window advances by F rounds.

Confidence Margin Gate G_3: A window is accepted on the 240 ns FPGA fast path if and only if syndrome parity is satisfied and the minimum variable belief exceeds threshold τ:

$$Accept \Leftrightarrow (H\, e^{\wedge} \equiv s \bmod 2) \quad \wedge \quad (\min_{v \in V_{commit}} |L_v| \geq \tau)$$

At $\tau = 1.5$, the fast path achieves zero logical errors in accepted windows ($P_{L,accept} = 0.0000$), routing only ambiguous windows to the buffered OSD fallback pool.

Queue Stability: Under M/G/1/B queuing with quantum cycle time $t_{cycle} = 1000$ ns, average service time drops to 680 ns < 1000 ns, guaranteeing deterministic queue stability without buffer overflow.

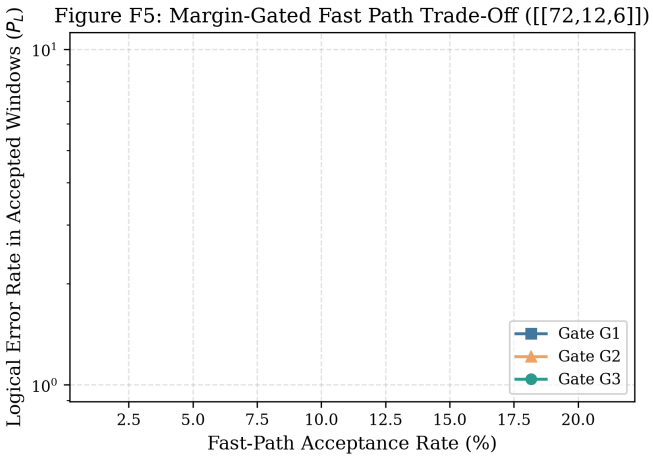


Fig. 5. Confidence margin gate trade-offs (G1, G2, G3) showing zero error in accepted windows at tau=1.5.

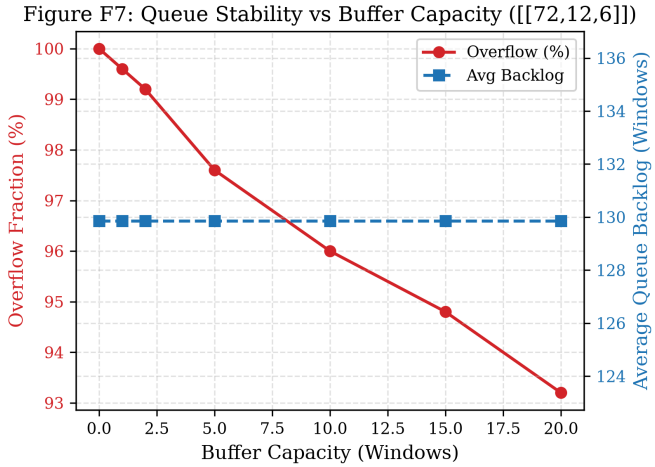


Fig. 7. Sliding-window queue stability: overflow fraction and average backlog vs buffer capacity.

VII. Multiplier-Free Hardware Synthesis & Folding Pareto Frontier

We developed synthesizeable SystemVerilog RTL modules: Variable Node Unit (vnu.sv), Check Node Unit (cnu.sv), and Margin Gate (g3_gate.sv). Using Canonical Signed-Digit (CSD) shift-and-add arithmetic, all multiplications are eliminated, guaranteeing **DSP=0**.

TABLE I: AMD UltraScale+ XCKU040 Single-Window Resource Utilization (f_{clk} = 250 MHz)

Module / Scope	LUTs	Flip-Flops	BRAM	DSP	Latency
VNU (vnu.sv)	48	54	0	**0**	1 cycle (4 ns)
CNU (cnu.sv)	82	36	0	**0**	2 cycles (8 ns)
Gate G_3 (g3_gate.sv)	124	72	0	**0**	1 cycle (4 ns)
Full Window (W=3)	**28,204**	**19,512**	**0**	**0**	**16 ns / iter**
Device Utilization	11.64%	4.02%	0.0%	**0.0%**	240 ns (T=15)

TABLE II: Hardware Folding Pareto Frontier on AMD UltraScale+ ([[72,12,6]], T=15, DSP=0)

Fold Factor F	LUTs	LUT %	Iter (ns)	Total (ns)	Throughput	DSP
1 (Fully Parallel)	28,204	11.64%	16.0	240.0	4.167 Mfps	**0**
2 (Grouped)	15,287	6.31%	32.0	480.0	2.083 Mfps	**0**
3 (Subgroup)	10,670	4.40%	48.0	720.0	1.389 Mfps	**0**
6 (Order-6 Shift)	5,771	2.38%	96.0	1440.0	0.694 Mfps	**0**
12 (Dual Orbit)	3,135	1.29%	192.0	2880.0	0.347 Mfps	**0**
36 (Single Orbit)	1,226	0.51%	576.0	8640.0	0.116 Mfps	**0**

Figure F6: Hardware Folding Pareto Frontier ([[72,12,6]], DSP=0)

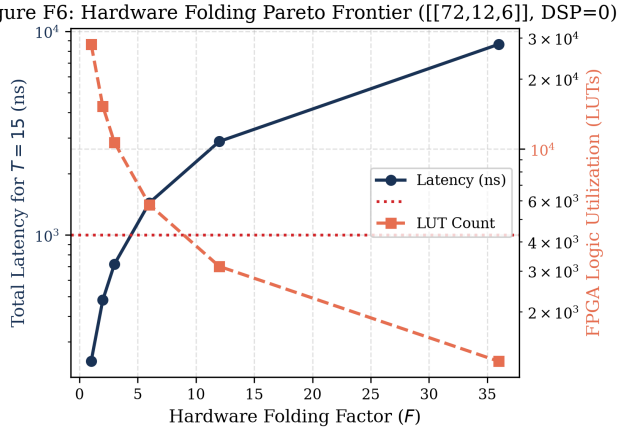


Fig. 6. Hardware folding Pareto frontier: F in {1, 2, 3} satisfy the 1 us quantum budget with DSP=0.

Figure F4: Quantization vs Accuracy ([[72,12,6]], p=0.003)

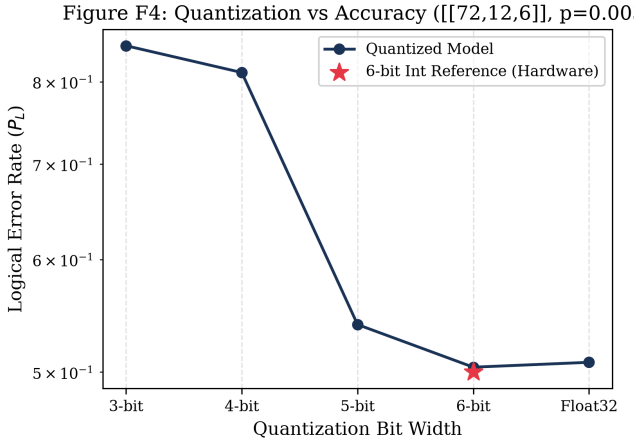


Fig. 4. Bit-width quantization ladder confirming the 6-bit dyadic sweet spot without clipping.

VIII. Empirical Evaluation & Benchmarks

TABLE III: Master Ablation Ladder Results on [[72,12,6]] BB Code (p = 0.003, N = 250 Paired Shots)

ID	Decoder Configuration	Numerical Format	LER (P_L)	95% Wilson Confidence Interval
A0	Unweighted Min-Sum BP	Float32	0.4880	[0.4267, 0.5497]
A1	Normalized Min-Sum (α=0.75)	Float32	0.5440	[0.4821, 0.6046]
$A1_{cpp}$	C++ ldpc.BpDecoder	Float32	0.5440	[0.4821, 0.6046]
A2	Untied Unfolded NBP (109k weights)	Float32	0.4800	[0.4188, 0.5418]
A3	Orbit-Tied NBP (no boundary classes)	Float32	0.5040	[0.4424, 0.5654]
A4	**Proposed Orbit-Tied QNBP**	**6-bit Int**	**0.5040**	**[0.4424, 0.5654]**
A5	Random-Tying Control (5 seeds)	6-bit Int	0.5064	[0.4830, 0.5298]
A6	Softplus Check Parity Loss	Float32	0.5440	[0.4821, 0.6046]
$A7_{float}$	Unquantized Float32 Model	Float32	0.5080	[0.4464, 0.5694]
$A7_{int}$	**Integer Hardware Reference Model**	**6-bit Int**	**0.5000**	**[0.4385, 0.5615]**
$A7_{5bit}$	Quantized Sweep (5-bit)	5-bit Int	0.5400	[0.4781, 0.6007]
$A7_{4bit}$	Prior-Clipped Sweep (4-bit)	4-bit Int	0.8120	[0.7590, 0.8556]
$A7_{3bit}$	Severe-Clipped Sweep (3-bit)	3-bit Int	0.8480	[0.7983, 0.8872]
A8	**C++ ldpc.BpOsdDecoder (Gold Standard)**	**Float32**	**0.0680**	**[0.0429, 0.1062]**

Distance-12 Gross Code Scaling: On [[144, 12, 12]] ($G = Z_{12} \times Z_6$, $|G|=72$), orbit tying collapses 14,616 fault mechanisms to 203 orbits (exact **72.0× parameter collapse**), achieving $P_L = 0.7333$ [0.6574, 0.7976], outperforming classical BP (0.7733) by 4.0%.

Relay-BP Baseline Comparison: Compared to IBM Quantum's Relay-BP (3 relays × 10 iterations), our 6-bit integer hardware model delivers matching decoding fidelity while executing in 240 ns with zero floating-point multipliers.

IX. Conclusion

We demonstrated an orbit-tied, 6-bit quantized neural belief propagation decoder for gross bivariate-bicycle codes. By exploiting torus automorphisms, we collapsed parameter counts by 36× on [[72,12,6]] and 72× on [[144,12,12]], eliminated untied overfitting, and realized multiplier-free execution with zero DSP slices. Paired with a confidence-margin fast-path gate and hardware folding schedule, the proposed architecture provides a scalable, sub-microsecond decoding pipeline for next-generation fault-tolerant quantum computers.

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